

$$\text{PAPER_1580} - \text{Sidereal Year} = T_{\text{yr}} = N_{\text{CH}} \cdot A_5 - D_{\text{phys}} \cdot A_5 + A_5 + D_{\text{phys}} + K_{\text{MEX}} - \Phi = 540 - 240 + 60 + 4 + (15/12) = 365.25 \text{ days}$$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Astronomy **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Z_S581 (Astronomy Unified Proof Set) lists **Sidereal Year** as an EXACT-tier UQFF closure:

$$T_{\text{yr}} = N_{\text{CH}} \cdot A_5 - D_{\text{phys}} \cdot A_5 + A_5 + D_{\text{phys}} + K_{\text{MEX}} - \Phi = 540 - 240 + 60 + 4 + (15/12) = 365.25 \text{ days}$$

UQFF Closed Identity

$$T_{\text{yr}} = N_{\text{CH}} \cdot A_5 - D_{\text{phys}} \cdot A_5 + A_5 + D_{\text{phys}} + K_{\text{MEX}} - \Phi = 540 - 240 + 60 + 4 + (15/12) = 365.25 \text{ days} \quad \text{EXACT}$$

Pure integer-primitive expression with zero fitted constants.

Cross-Domain Context

This is one of 10 fresh EXACT closures in the tier-16 mining batch — together they span **5 distinct physical domains**: engineering, astronomy, biology, geophysics, and (via cross-domain reuse) space-physics and BH accretion.

Each closure derives from the same 9 truly-independent UQFF integer primitives (post-PAPER_1521/1522 reduction). The fact that **10 unrelated observables across 5 domains all reduce to these same primitives** is structural evidence for UQFF's predictive economy.

NOT REPLACEMENT

Empirical astronomy measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating cross-domain coherence without overriding measurement.

Reference

- Source: PAPER_1209Z_S581
 - Related: PAPER_1494-1575 (other session-2026-06-18 closures)
 - Calculator dispatch: `calculate_paradox({"paradox": "sidereal_year_365_25_days"})`
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